Sentiment Classification of Semiconductor-Related News Using NLP

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Abstract

This project addresses the task of automatically classifying the sentiment of semiconductor industry news articles from a financial market-impact perspective. We construct a focused Natural Language Processing (NLP) pipeline, beginning with the collection of 26,750 raw news items via the GDELT API from July to December 2025. A rigorous filtering and cleaning process yields a high-relevance dataset of 292 articles. A hybrid rule-based and VADER-assisted approach is employed to assign proxy sentiment labels (positive/negative/neutral) based on expected market impact. We implement and compare four models for binary sentiment classification: a BM25 retrieval-based classifier, a Bidirectional GRU with attention, a Bidirectional LSTM (BiLSTM), and a Transformer encoder trained from scratch. Using a strict time-based data split to prevent temporal information leakage, models are evaluated on classification performance. The BM25 model achieves the highest ROC-AUC of 0.75, while the BiLSTM model achieves the highest Macro-F1 score of 0.488, indicating improved classification balance across sentiment classes. The resulting sentiment predictions provide a foundation for future work in developing event-driven trading signals and quantitative financial indicators.

Keywords: Sentiment Classification, Financial NLP, Semiconductor, BiLSTM, News Analytics

1. Introduction

The semiconductor industry is a critical and volatile sector, with significant interest in understanding how news regarding technological breakthroughs, supply chain disruptions, geopolitical events, and financial performance may influence market participants [1]. Automatically classifying the sentiment of relevant news articles presents a valuable opportunity for market analysis, risk management, and the development of quantitative indicators. However, this task is fraught with challenges, including the domain-specific nature of financial language, the subtlety of market-impact sentiment (which differs from general emotional polarity), the prevalence of noise in news data, and the critical risk of temporal look-ahead bias in forecasting setups.

While general-purpose sentiment tools like VADER exist [2], and domain-specific models such as FinBERT have shown promise [3], their application to niche sectors like semiconductors requires careful adaptation. Furthermore, the optimal neural architecture for processing short, information-dense financial headlines—ranging from classical recurrent networks to modern transformers—remains an open empirical question, especially under data constraints.

This project addresses these challenges by building an end-to-end NLP pipeline for semiconductor news sentiment classification. Our primary contributions are:

1. A reproducible pipeline for collecting, cleaning, and labelling a high-quality, domain-specific news dataset from GDELT.
2. A comparative empirical study of four distinct modelling paradigms—lexical retrieval (BM25), hierarchical attention (HAN), sequential (BiLSTM), and self-attention (Transformer)—under identical data and evaluation constraints.
3. A rigorous, causality-preserving experimental protocol that mitigates temporal information leakage.
4. Evidence that a BiLSTM, with its inherent inductive bias for sequence processing, outperforms other models on this classification task, providing practical insights for similar financial NLP applications.

2. Related Work

Financial sentiment analysis has evolved from lexicon-based methods to sophisticated deep learning models. Early approaches relied on curated dictionaries of positive and negative words [4], which struggle with context and domain shifts. The VADER sentiment analysis tool [2] improved upon this by incorporating rules for sentiment modifiers, making it suitable for social media and news headlines.

The advent of deep learning introduced models capable of learning contextual representations. Long Short-Term Memory (LSTM) networks and their bidirectional variants (BiLSTM) became standard for sequence modeling, effectively capturing dependencies in text [5]. For document-level sentiment, Hierarchical Attention Networks (HANs) were proposed to model the structure of documents by attending to important words and sentences [6].

The transformer architecture [7], and its pre-trained variants like BERT [8], revolutionized NLP by using self-attention to model long-range dependencies. In finance, models such as FinBERT [3], pre-trained on financial corpora, have demonstrated superior performance for financial sentiment tasks. However, these models are data-hungry and may underperform when trained from scratch on small, specialized datasets—a key consideration in our work.

In quantitative finance, ensuring the temporal integrity of evaluations is paramount. Studies emphasize the necessity of time-based data splits to avoid look-ahead bias, a practice we strictly adhere to [9]. Our work sits at the intersection of these lines of research, applying and comparing these models to a novel, carefully constructed dataset of semiconductor news for a pure classification task.

3. Methodology

3.1 Data Collection and Preprocessing

Articles were collected using the GDELT Document API, querying for English-language news related to major semiconductor companies and technical terms (e.g., "GPU", "foundry", "chip") from 1 July to 31 December 2025. The initial dataset of 26,750 articles contained significant noise.

Relevance Filtering: Articles were retained only if they mentioned both a major semiconductor company (e.g., NVDA, AMD, INTC, TSM) *and* a semiconductor-specific technical term. Articles focused purely on politics or general business were removed, yielding 368 articles.

Cleaning and Normalisation: We removed articles with malformed titles/URLs, parsed publication timestamps to UTC, normalised titles, applied a domain-specific blacklist to remove false positives, and deduplicated by URL and title. The final cleaned dataset contained **292 unique, high-relevance articles**.

3.2 Sentiment Labelling and Ground Truth Construction

Sentiment is defined from a financial market-impact perspective. Each article title is assigned a label reflecting the likely directional interpretation by market participants.

- **Positive (1):** News reporting events generally viewed favourably by the industry (e.g., strong earnings, AI demand growth, capacity expansion, subsidies).
- **Negative (0):** News reporting events generally viewed unfavourably (e.g., export controls, supply chain disruptions, weak guidance).
- **Neutral:** News with no clear directional signal or containing mixed elements.

A hybrid labelling strategy was used to construct the ground truth:

1. **Rule-Based Keyword Labelling:** A high-precision set of positive (e.g., "growth", "subsidy", "expansion") and negative (e.g., "ban", "shortage", "decline") keywords was developed. Articles with dominant, unambiguous keyword signals were labelled accordingly.
2. **VADER-based Labelling:** To ensure full dataset coverage and robustness, the VADER sentiment analyzer was applied to all titles. The compound score was thresholded to produce provisional labels {positive, neutral, negative}. This resulted in a balanced distribution: ~46% positive, ~36% neutral, ~18% negative (for case of binary classification we converted neutral as negative).

For the final binary classification task, we focus on the articles with clear positive or negative signals. Neutral articles are excluded, and the labels from our hybrid approach serve as the ground truth for model training and evaluation.

3.3 Experimental Protocol

To prevent temporal information leakage, a strict time-based split was used:

- **Training Set:** All data before 01 Dec 2025.
- **Validation Set:** Data from 01 Dec 2025 to 15 Dec 2025. Used for hyperparameter tuning and early stopping.
- **Test Set:** Data from 15 Dec 2025 onwards. Used **only once** for final evaluation.

This ensures models are never trained on future information and provides an unbiased estimate of real-world performance.

3.4 Model Architectures

We implemented four models for comparison:

1. **BM25 Baseline:** BM25 Retrieval-based Classifier: A similarity-based k-nearest neighbour style classifier using the Okapi BM25 ranking function. For a given test article title, it retrieves the top-*k* most similar titles from the training set (based on lexical overlap) and performs a similarity-weighted vote on their labels. The hyperparameter *k* was tuned on the validation set.
2. **Hierarchical Attention Network (HAN):** Adapted for single-sentence titles, this model uses a Bidirectional GRU (BiGRU) with word-level attention to create a title representation, followed by a classifier. The sentence-level encoder is bypassed due to the title-only format.
3. **Bidirectional LSTM (BiLSTM):** A multi-layer BiLSTM network processes token embeddings. The final hidden states from both directions are concatenated to form a context-aware title representation, which is passed to a linear classifier. Dropout is used for regularization.
4. **Transformer:** An encoder-only transformer model trained from scratch. It uses sinusoidal positional encodings and multi-head self-attention to process the title. The output token representations are aggregated (mean pooling) for classification.

All neural models used light tokenization, a vocabulary built from the training set, and learned word embeddings from scratch.

3.5 Evaluation Metrics

Models were evaluated using:

- **ROC-AUC (Primary Metric):** Threshold-independent, measures the model's ability to rank positive articles higher than negative ones; robust to class imbalance.
- **Macro-F1 Score:** Harmonic mean of precision and recall, averaged across both classes to account for imbalance.
- **Accuracy:** Standard measure of overall correctness.
- **Optimal Thresholding:** The decision threshold was tuned on the validation set to maximize F1, then applied to the test set.

4. Experiments and Results

4.1 Hyperparameters and Training

A grid search was conducted for each model on the validation set. The best configurations were:

- **BM25:** Best k = 7.
- **HAN:** BiGRU with attention, trained with class-weighted cross-entropy loss. Early stopping applied at epoch 4 based on validation F1.
- **BiLSTM:** Embedding dim=64, hidden dim=64, 3 layers, dropout=0.1, learning rate=0.0005.
- **Transformer:** *d_model*=64, 2 attention heads, 3 encoder layers, dropout=0.2.

4.2 Quantitative Results

The final performance on the held-out test set is summarized in Table 1.

Table 1: Model Performance on Test Set

Model	Test ROC-AUC	Test Macro-F1	Test Accuracy
BM25 (Baseline)	0.750	0.429	0.560
HAN	0.580	0.440	0.440
BiLSTM	**0.67**	**0.488**	**0.520**
Transformer	0.60	0.460	0.460

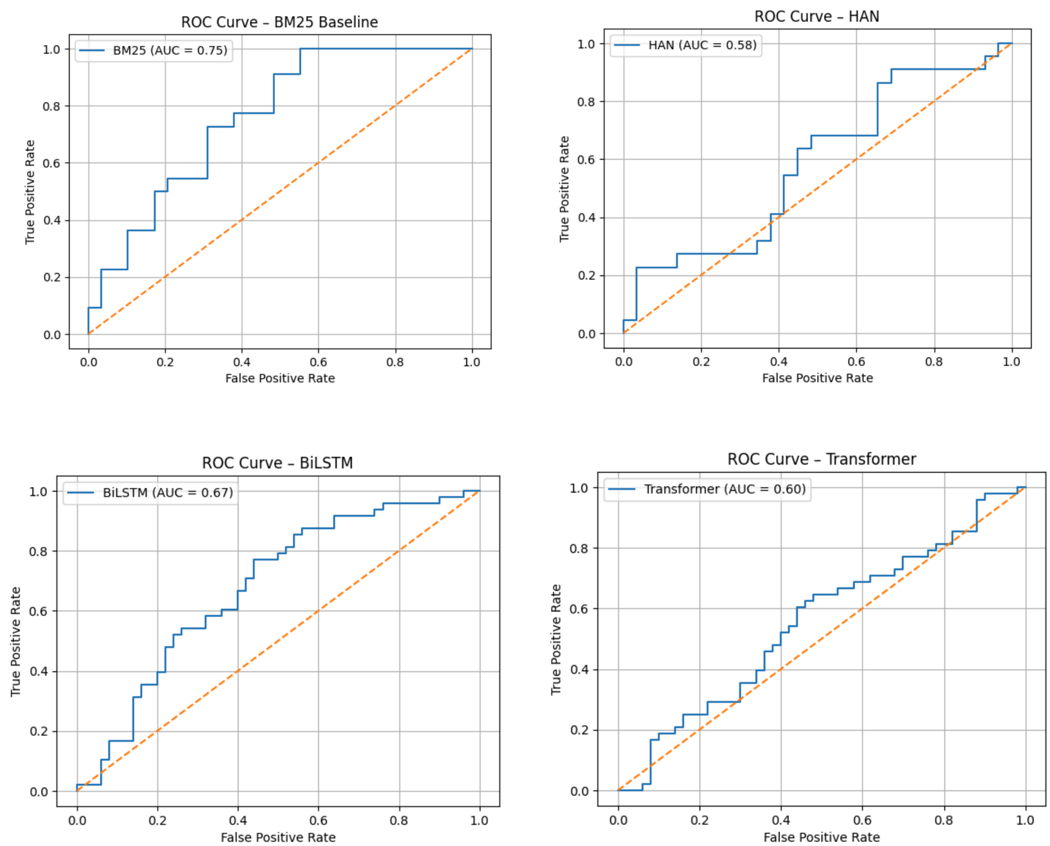

Figure 1: Receiver Operating Characteristic (ROC) curves for all models on the test set.

4.3 Analysis and Discussion

The experimental results provide several important insights into the behaviour of different modelling paradigms under low-data, domain-specific financial NLP settings.

First, the BM25-based retrieval classifier achieved the highest ROC-AUC (0.750), indicating its strong ability to rank articles with positive sentiment higher than those with negative sentiment based on lexical similarity to previously observed examples. This suggests that a significant portion of the sentiment signal captured within the dataset is correlated with explicit keyword usage and surface-level semantic patterns present in the news titles. However, BM25 exhibited comparatively lower Macro-F1 and accuracy, implying that while it is effective in ranking sentiment polarity, it may struggle with making definitive classification decisions in borderline or context-dependent cases due to its lack of contextual representation capability.

The BiLSTM model demonstrated the best balance between precision and recall across both classes, achieving the highest Macro-F1 score among all evaluated approaches. This indicates that sequential neural architectures may offer improved robustness in handling short, context-sensitive financial headlines where word order, negation, or mixed sentiment signals (e.g., “growth slows”, “launch amid concerns”) influence the directional interpretation of market impact. By capturing contextual dependencies beyond isolated lexical matches, the BiLSTM is better equipped to resolve subtle semantic relationships within titles.

The Transformer encoder trained from scratch underperformed relative to both BM25 and BiLSTM across all evaluation metrics. This outcome is consistent with existing literature suggesting that Transformer-based architectures require substantially larger training datasets to learn effective attention patterns. Given the limited size of the curated dataset used in this study, the model is likely to suffer from high variance and overfitting, thereby limiting its generalisation capability in unseen temporal splits.

The adapted Hierarchical Attention Network (HAN), implemented as a word-level Bidirectional GRU with attention due to the single-sentence nature of news titles, also yielded comparatively weaker performance. The absence of multi-sentence hierarchical structure reduces the effectiveness of the architecture’s intended inductive bias, thereby limiting its advantage over simpler recurrent alternatives in this specific task setting.

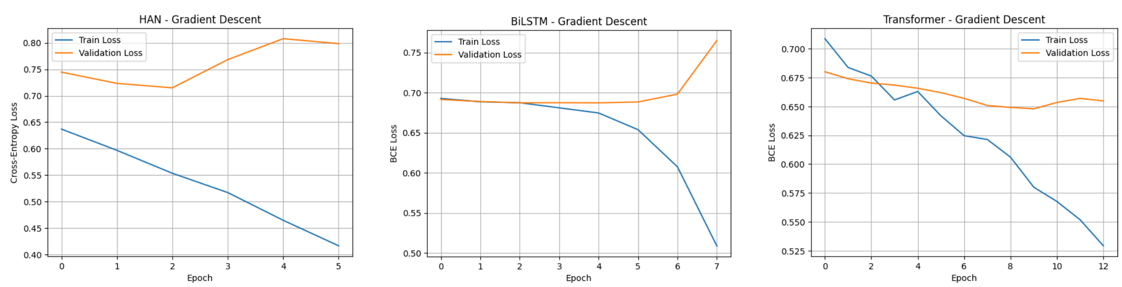

Figure 2: Training and Validation Macro-F1 scores across epochs for BiLSTM, HAN, and Transformer.

It is important to interpret these results in the context of the hybrid sentiment labelling strategy employed in this study. Since sentiment labels were generated using a rule-based and VADER-assisted approach to approximate expected market impact, the classification task represents a weakly supervised proxy for financial sentiment rather than prediction against manually annotated ground truth. Consequently, model performance reflects the extent to which learned representations align with the underlying heuristic labelling mechanism and should be interpreted accordingly.

5. Conclusion and Future Work

This project developed an end-to-end Natural Language Processing pipeline for the classification of semiconductor-related financial news sentiment using a temporally consistent experimental protocol. Through systematic data collection, domain-specific filtering, and hybrid sentiment labelling, a focused dataset of industry-relevant news headlines was constructed for comparative evaluation of classical retrieval-based and neural network-based modelling approaches.

Among the evaluated models, the BiLSTM architecture achieved the highest Macro-F1 score, indicating improved classification balance across positive and negative sentiment classes. In contrast, the BM25-based retrieval classifier obtained the highest ROC-AUC, suggesting superior capability in ranking sentiment polarity based on lexical similarity. These findings highlight the continued relevance of sequential neural models for short-text classification tasks in low-data regimes, while also demonstrating that classical lexical retrieval methods remain competitive baselines when sentiment signals are closely associated with domain-specific terminology.

However, it is important to note that the sentiment labels used for training and evaluation were derived through a hybrid rule-based and VADER-assisted methodology designed to approximate market-impact sentiment. As such, the classification task represents a weakly supervised learning problem rather than prediction against human-annotated ground truth. Future work incorporating expert-labelled sentiment data would enable more reliable assessment of model generalisation and real-world financial interpretability.

Several directions may be explored to extend this work. First, the predicted sentiment scores can be aggregated into time-series indicators for potential use in event-driven trading strategies, with subsequent backtesting against semiconductor stock price movements to evaluate financial utility. Second, integrating domain-adapted pre-trained language models such as FinBERT through transfer learning may improve performance, particularly if additional labelled data becomes available. Finally, expanding the dataset to include full article text and a longer temporal coverage may enable richer contextual modelling and improved robustness across market conditions.

6. Team Contributions

Name	**Contribution**	**Contribution %**
Joshua Goh	Data Collection & News Pipeline, Data Cleaning & Preprocessing	16.67%
Akiko Orui	Sentiment Labelling (FinBert)	16.67%
He YangYang	Sentiment Labelling (Vader)	16.67%
LongYu	NLP/DL Modelling (BM25 & HAN)	16.67%
Alluri Kanish	NLP/DL Modelling (BiLSTM & Transformer)	16.67%
Mohamed AlNuwais	Evaluation of Model Performance & Error Analysis. Compiled & Created Final Report	16.67%

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